



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Introduction

TURTLE Hatchling

What is Hatchling?



An introduction to foundational technical, soft, and engineering skills that culminates in an internal competition project



Historic Success



Hatchling formed in Fall 2015.



Over the past two years, participant roles currently include:

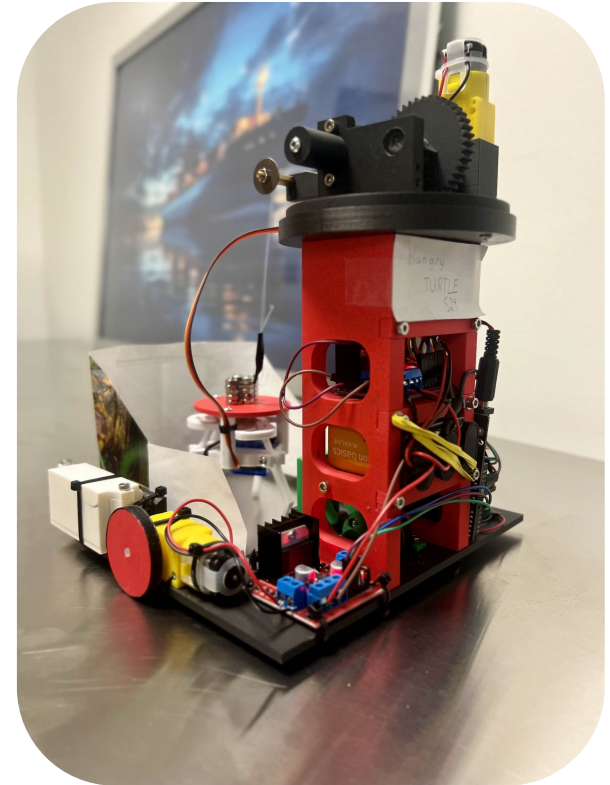
- Engineering student org officers
- Project team leads
- Project sub-team leads
- Solidworks certified (CSWA and CSWP)
- Research and Internship positions

All within two semesters of completing Hatchling!!

It is an OPPORTUNITY to ...



- Learn about engineering
- Get hands-on experience
- Showcase Abilities
- Develop...
 - Technical Skills
 - Critical Thinking
 - Problem Solving
 - Friendships





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Overview

Learning Objectives



- **SolidWorks (CAD) Competency :**
 - Design custom parts and modify assemblies
 - Reading and designing around manufacturing constraints
- **Electronics**
 - Evaluating hardware specifications and datasheets
 - Circuit design and implementation
- **Programming**
 - Utilize Git/GitHub
 - Control hardware via a microcontroller
 - Python and C++
- **Manufacturing**
 - Soldering and crimping
 - Additive manufacturing
- **Problem Solving and Critical Thinking**
 - Teams go through the design process from concept creation to testing



Schedule



Week 1: Introduction

Week 2: SolidWorks (CAD) Foundation

Week 3: SolidWorks 3D

Week 4: Tools and Process

Week 5: Design Review and C++

Week 6: SolidWorks Assembly

Week 7: Programming

Week 8: Electronics and Soldering

Week 9: Prototype Week

Week 10: Build Week

3 meetings a week

- Tuesday: 6 - 8 PM
- Thursday: 4 - 6 PM
- Friday: 5 - 7 PM

Attend at least one meeting a week. If there is a conflict, let an officer know.

Information will be the same across meetings, however, feel free to attend more than one. Slides are posted on the website.

Note: Weeks with project milestones are in **orange**

Doing some math



Hatchling Directors are volunteers. What we get out of this is the ability to meet y'all and help you become the most knowledgeable version of yourself. Unfortunately **time** is the limiting factor.

Meeting hours = 10 weeks * 2 hours = 20 hours

Tip: To get the most out of this opportunity, dedicate an extra 1-2 hours a week to learn more about the topic.

Oh and listen to the old phrase “Ask questions”

Another Note: TURTLE



- Our capacity for Hatchling is 120 people
 - We do **NOT** allow for former Hatchlings to repeat Hatchling
- Advanced Projects are not guaranteed for Hatchling graduates
 - Every Advanced Project lead determines their own team
 - Selection is based on the needs of the project
 - Most will look for the Hatchling project on your resume
 - Do NOT lie about Hatchling
 - A lot will check if you finished a Hatchling robot
 - If you are interested in a certain project, contact the project lead and learn more about what skills they are looking for

General Lab Rules



- An officer must be present in the lab at all times while the lab is in use.
- Any behavior deemed disrespectful or inappropriate can result in disciplinary action or a request to vacate
- During a project's reserved lab time, the project lead may ask anyone to leave the lab for any reason.
- Project tasks always have priority over personal or school projects when it comes to lab tools and space.
- **Misuse or unsafe lab practice will result in the revoking of privileges.**
- If you need training or supervision for a tool and cannot receive it from those present, send an @Officer in the
- If at any time, common sense would defy any of the rules detailed in this document, ask an officer for help.

Note: Refer to your organization's rules

General Lab Rules



While in the lab, your safety is first and foremost your own responsibility. You are surrounded by resources and experienced people who can help you use them. If you are at any point unsure of what you are doing, ask. If you feel unsafe with a tool, stop what you are doing immediately and do not resume work until you have received clarification.

Tools Requiring Training:

- Angle Grinder
- Dremel
- Drill Press
- Hand Drill
- Impact Driver
- Jig Saw
- Power Supply
- Printer (3D)
- Soldering Iron

Tools that require Supervision:

- Angle Grinder
- Dremel

Safety Gear

- PPE is required for all tools listed in the training section excluding the 3D printers. Proper PPE includes: closed toe shoes, bottoms that cover to the lower leg, a sleeved shirt, and safety glasses. All loose items and hair must also be properly secured up and out of the way.

Trainings

- Any individual may be trained by an officer. This training is tool specific and is not all encompassing.



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Officer / Org Introductions

TURTLE

Texas A&M University Robotics Team and Leadership Experience



The oldest active robotics org at TAMU

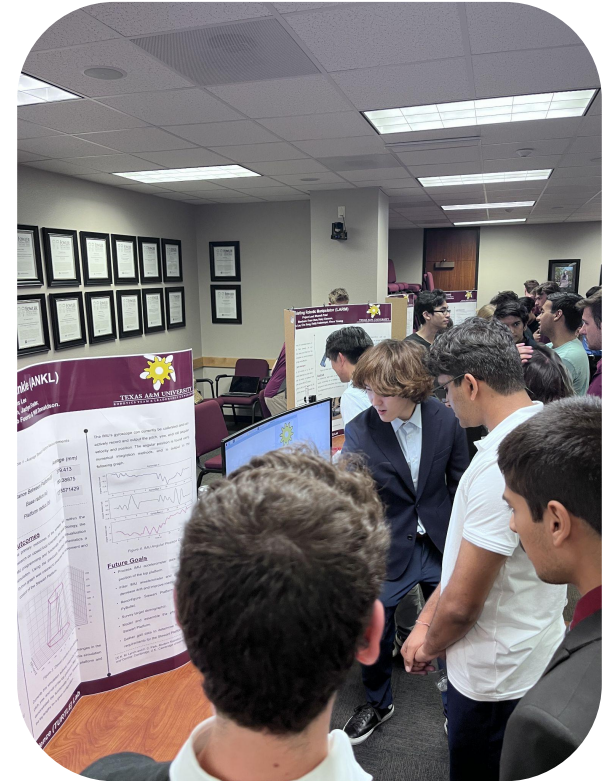
Founded in 2013

400+ Active members

19 Student Led Projects

Mission:

To develop the next generation of leaders through hands on robotics projects, competitions, and community service opportunities.



Isaac Cabello - Hatchling Chair



Email: isaac.cabello@tamu.edu

Discord: Chicknwing



- MEEN 2027
- Meeting Days: TBD
- Experience:
 - Former Hatchling!
 - POBS Project lead
 - LANL Student Researcher
 - Autonomous Systems Lab
 - Bio-Inspired Systems Lab
- Fun facts
 - I enjoy yapping
 - I am the second greatest Hatchling director of all time

Julia Sopala - Development VP



Email: jusopala18@tamu.edu

Discord: juli.a9847



- BMEN 2027
- Meeting Days: TBD
- Experience:
 - OLSN
 - R&D Engineering Intern
 - Biomedical Research
 - Hatchling Instructor
 - Former Hatchling
- Fun facts:
 - I want to be fluent in 4 languages
 - I'm 3/5th of the way there
 - I want to learn how to surf



Ethan Real - Hatchling/MI Instructor



Email:ethanreal05@tamu.edu

Discord: ethan_real05



- MEEN 2027
- Meeting Days: Mondays
- Experience:
 - Turtle
 - Former Hatchling Member
 - BEEST
 - Interned @ Sulzer
- Fun facts:
 - I'm a Fraternal Twin
 - I love to go fishing whenever I can

Santiago Yordi - Hatchling Instructor



Email: santiyordi@tamu.edu
Discord: santiyordi



- MEEN 2028
- Meeting Days:
- Experience:
 - EDEN
 - Former Hatchling
 - ASME Go Kart
- Fun facts:
 - I love baseball (Braves)
 - I love reading
 - ASOIAF and Dune especially

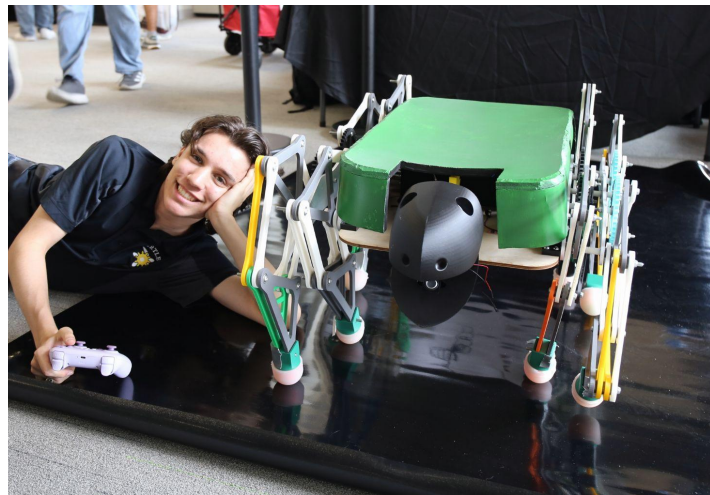


Lorenzo Hidalgo - Hatchling/MI Instructor

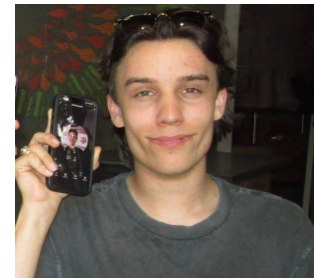


Email: lhidalgo@tamu.edu

Discord: darthzo



- MEEN 2027
- Meeting Days
 - TBD
- Experience:
 - TURTLE
 - BEEST
 - SHDR
 - OLSN
- Fun facts
 - I've had white hair since I was born
 - I've had two family disappearances



Kayla Sam - Hatchling/SI Instructor



Email: kaysam@tamu.edu
Discord: kkaayyss



- ENGR 2029
- Meeting Days: TBD
- Experience:
 - FASH Electrical Team Lead
 - FTC Robotics
- Fun facts:
 - I have a twin brother
 - I sing in an A&M Choir
 - I'm living in a temporary apartment 🌹



Andrew Sun - Hatchling Instructor



Email: andrewsun.15@tamu.edu

Discord: goatedogger



- MMET 2029
- Meeting Days: Friday
- Experience:
 - Former Hatchling
 - CMBT
 - Aggies Create
- Fun facts:
 - I'm friends with the goat
 - Tennis
 - Basketball
 - Severance & AOT are my fav show of all time

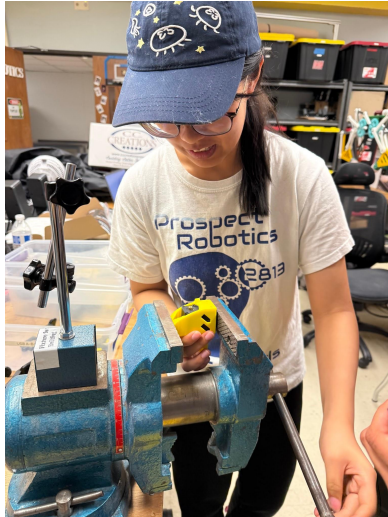


Grace Wang - Hatchling Instructor



Email: wanggrace@tamu.edu

Discord: .waku.waku.



- MEEN 2029
- Meeting Days: TBD
- Experience:
 - OLSN
 - ACHE
 - FRC
- Fun facts:
 - I'm not from Texas
 - I have 2 pet ducks

Mohid Ismail - Hatchling Instructor



Mohid Ismail

Email: mohidismail@tamu.edu

Discord: bean.man.



- ELEN 2027
 - Electrical Engineering
- Meeting Days:
 - Tuesdays
 - Thursdays
- Experience
 - TURTLE
 - VIRT
 - Former Hatchling member
- Fun Facts
 - Lost a 3v1 to a goat once
 - Once played terraria for 18 hours straight

William Shan - Hatchling Instructor



William Shan

Email: williamshan698@tamu.edu

Discord: [williamthepig1](#)



- MEEN 2028
 - Mechanical Engineering
- Meeting Days:
 - Fridays
- Experience
 - Aggies Create
 - EcoFil
 - TURTLE
 - Former Hatchling
 - CMBT
 - Interned @ Flowserve
- Fun Facts
 - TURTLE Smash Bros Ultimate Champion
 - Lost by 70 points in Basketball IM's...



Khanathan Vo - Hatchling/MI Instructor



Email: Khanh0315@tamu.edu

Discord: M1necrftpr0



- MEEN 2028
- Meeting Days:
 - None
- Experience
 - TURTLE
 - BLNC
 - POBS
 - Hatchling Fall 2025 winner
 - HERC Lab OSL
 - Doordash Driver
- Fun Facts
 - I like skating and running and volleyball (larp)
 - I like sleeping in the lab or scc

McKenzie McCain - Product Manager



Email: kemping01@tamu.edu
Discord: kemping01



- ITDE 2027
 - Interdisciplinary Engineering (Biomechanics & Robotics Track)
- Meeting Days:
 - Fridays 4-6
- Experience
 - TURTLE
 - Former Hatchling
 - OLSN
 - CtrlRobot Lab
- Fun Facts
 - I know the 100 digits of pi
 - My initials are $E=(Mc)^2$
 - I drive a Camaro



Thomas Lopez - Hatchling Instructor



Email: thomaslopez@tamu.edu

Discord: senorsombero

- MEEN 2028
- Meeting Days: Tuesday/Thursday
- Experience:
 - TURTLE
 - OLSN (Mechanical Lead)
 - SHDR
- Fun facts
 - I have kissed multiple sharks
 - I will teach you how to lasso and Two-Step



Nathan Mersino - Hatchling Instructor



Nathan Mersino

Email: nathanmersino@tamu.edu

Discord: nathanmersino



- ELEN 2028
 - Electrical Engineering
- Meeting Days:
 - TBD
- Experience
 - TURTLE
 - POBS
 - Former Hatchling Member
- Fun Facts
 - 1220 Day Duolingo Streak
 - Photography
 - I am a Steelers fan and watch the team go 9-8 or be first round exits every year :(



Karthik Jayakumar - Hatchling Instructor



Karthik Jayakumar

Email: karthikjayakumar@tamu.edu

Discord: kjbossmen



- AERO 2028
 - Aerospace Engineering
- Meeting Days:
 - Friday
- Experience
 - SAE Aero Design
 - Nanostructured Materials Lab
 - TURTLE
 - Former Hatchling
- Fun Facts
 - I drive stick

John Neal - Hatchling Instructor



John Neal

Email: johnpneal@tamu.edu

Discord: wrook_



- MEEN 2028
 - Mechanical Engineering
- Meeting Days:
 - Friday
- Experience
 - Hatchling Grad!
 - LARM (Discontinued TURTLE Project)
 - Dr. Friesen's Haptics Lab
 - Joining CNTR this semester
- Fun Facts
 - I don't drive stick
 - President of the Fencing Club! (hmu if you want to join)

SOMTECH

Society of Mechatronics Engineers



Founded in 2020, currently 80 members!

We are a professional and competitive organization!

- University Rover Challenge (URC) : Building a prototype Mars rover to compete against international teams at the Mars Desert Research Station (located in Hanksville, Utah).

SOMTECH was founded as the Society of Mechatronics Engineers. Over the years, we have grown and expanded our group to over a hundred members of all engineering disciplines to help achieve our ultimate goals. Our mission is to include everyone from all backgrounds. We aim to empower individuals by providing equal opportunity and accepting all. If you have the passion and dedication, you're in!



Erin Galicha

Email: eringalicha@tamu.edu

Discord: enxgsx



MXET 2027 (Whoop!!)

Multidisciplinary Engineering Technology - Mechatronics

Meeting Days: Tuesdays and Fridays!

Experience

- Aggie Lemon Racing
 - Powertrain Team (I make car go BrAMMM Brrrahmm
Brahm Ram aarammm)
 - Design Team
- L3Harris Intern
- I made M-O from WALL-E!!!

Fun Facts

- I was classically trained in Opera
- I love spiders
- I sell paintings / am a brand designer
(I really like painting sad clowns)



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Icebreakers!

Split Up!



Form groups of 4 people. Duo up within that. Share your hot takes with each other.



“I WOKE UP FEELING THE CHEESIEST COACH!”

“I WOKE UP FEELING THE CHEESIEST COACH!”

Discuss



In your duo, pick one of your hot takes for the opposing duo to defend against you.

If y'all have bad ideas use these:

- Pineapple **SHOULD** be on pizza! - Julia
- A hotdog **IS** a sandwich! - Isaac
- Night showers beat morning showers! - William



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Program Installation yay!

Download the slides at:

turtlerobotics.org/hatchling

*Always recommend bringing a mouse and your laptop charger



Two options

*Mac users

Texas A&M students have free access to SolidWorks Student Premium through the Software Center. **Recommended**

<https://software.tamu.edu/>

DO NOT
INSTALL IN
ONEDRIVE

DO NOT
INSTALL IN
ONEDRIVE

The software can also be accessed through the Virtual Open Access Lab (VOAL) which can be used without downloading SolidWorks.

For Mac users [Omissa
Horizon Clients]

and

People who are currently
installing SolidWorks

<https://voal.tamu.edu/>

Note: SolidWorks is the recommended software for TURTLE Advanced Projects. Please try to utilize SolidWorks, however, other CAD platforms may be accommodated.

Bambu Labs Slicer



Slicer programs for 3D-Printing all function very similarly (Most are forks of Slic3r, Cura, or Simplify3D). This setup will depend on the brand, type of printers, and filaments available

Printer Selection

 Bambu Lab X1 Carbon ☒ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab X1 ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab X1E ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab P1P ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle
 Bambu Lab P1S ☒ 0.4mm nozzle ☐ 0.2mm nozzle ☒ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab A1 mini ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	 Bambu Lab A1 ☐ 0.4mm nozzle ☐ 0.2mm nozzle ☐ 0.6mm nozzle ☐ 0.8mm nozzle	

Most common filaments are:

1. PLA - Polylactic acid
2. PETG - Polyethylene terephthalate glycol

<https://bambulab.com/en-us/download/studio>

VS Code



Integrated Development Environment (IDE)

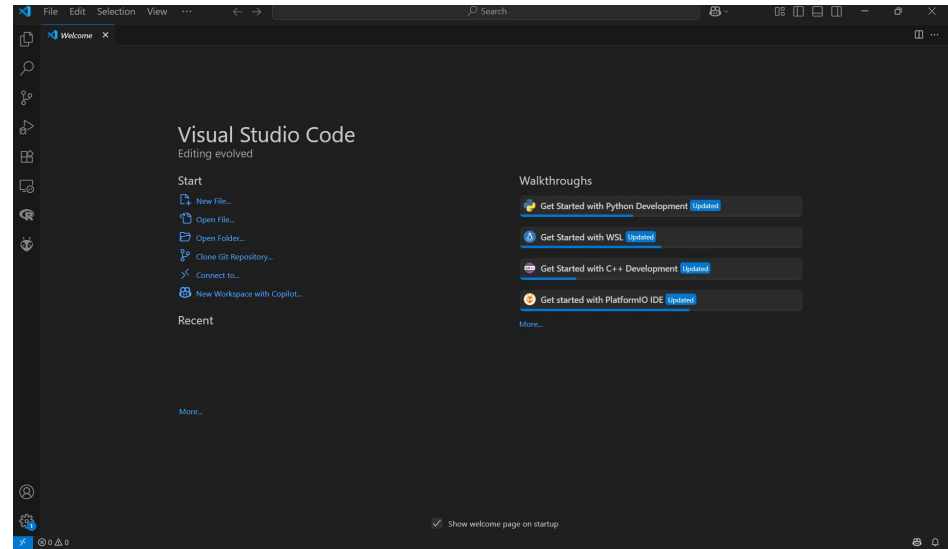
Powerful, popular, and customizable

Git and GitHub integration

Versatile across languages and types of project

Microsoft Store or

<https://code.visualstudio.com/>



Extensions to add

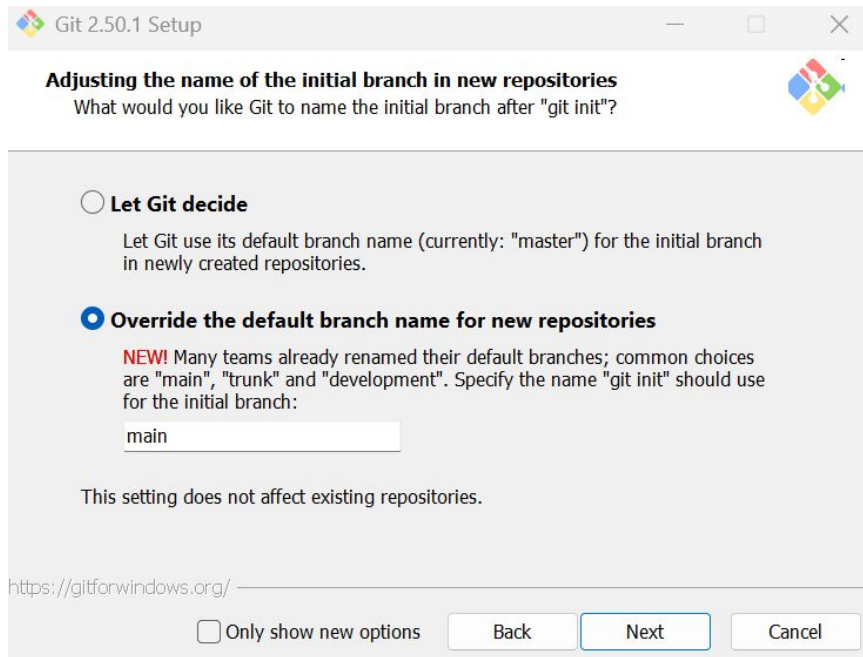
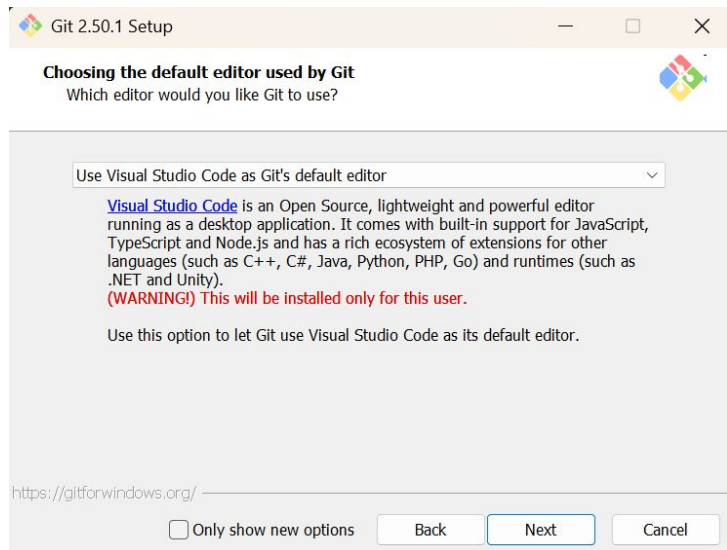


- C/C++ Extension Pack by Microsoft
 - Adds syntax highlighting
- PlatformIO IDE
 - Allows interfacing with ESP32
- Prettier (Personal preference)
 - Code formatter that makes code more legible
- GitHub CoPilot (Be careful to avoid using this for class)
 - AI chat and coding assist that scans files and autocompletes

Installing Git



- Go to this website to install Git:
 - <https://git-scm.com/downloads>
- Use the defaults on every step except the name of the “master” branch. Change the name to “main”



Github Account and Git Setup



1. Create a GitHub account

<https://github.com/>

2. Configure Git:

Open a terminal (or command prompt)

Set your username: `git config --global user.name "<First Last Name>"`

Set your email address: `git config --global user.email "<your@email.com>"`

3. Create a Repository on GitHub:

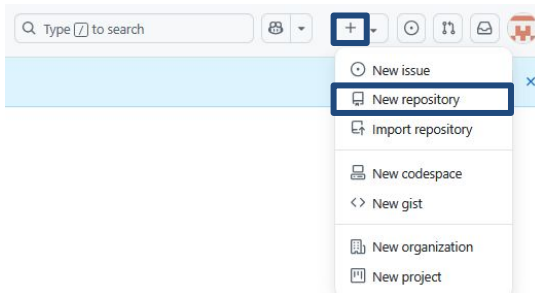
Click the "+" icon in the top right corner and select "New repository"

Give your repository a name and an optional description

Choose a repository visibility (public or private)

Click "Create repository"

Note: Hatchling uses phrases in angled brackets `<>` as placeholders. I.e. someone named John Doe would put "John Doe" instead of "`<name>`"

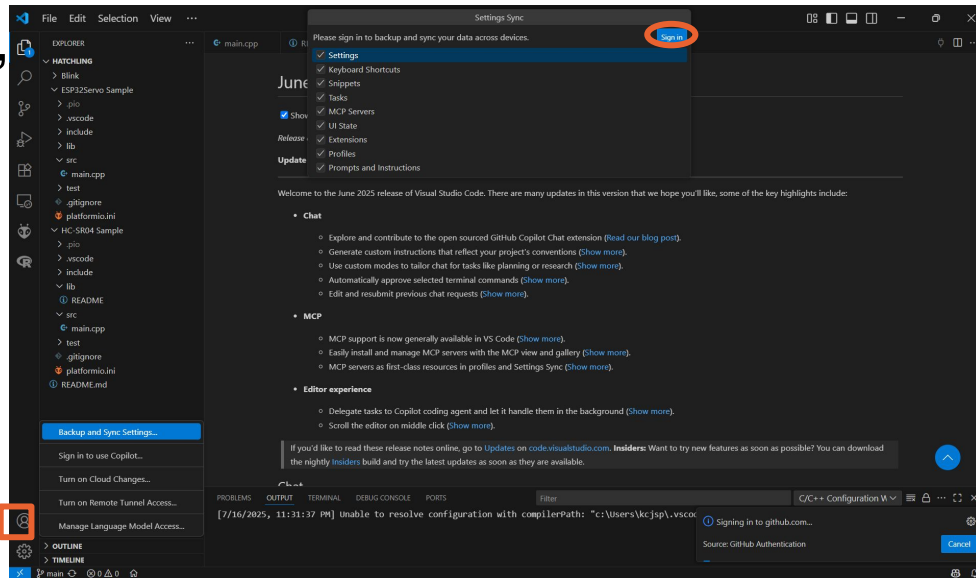


Adding GitHub to VS Code



1. Click “Accounts”
2. Click “Backup and Sync Settings”
3. Click “Sign In”
4. Click “Sign in with GitHub”
5. Authorize account access
6. Confirm by clicking “Accounts”

Note: Git is required to be installed. Git commands will work in the integrated terminal



ESP32 Communication Driver



Silicon Labs CP210x USB to UART Bridge must be installed for your laptop to communicate with the ESP32

Linux: `sudo adduser <username> dialout`

Windows: Google "[CP210x](#)"

Most will download "CP210x Windows Drivers" and run "CP210xVCPInstaller_x64.exe"

Software Links



- SolidWorks - (Do **NOT** install in OneDrive)
<https://software.tamu.edu/>
- Visual Studio Code [VS Code] - (Also on Microsoft Store)
<https://code.visualstudio.com/>
 - Extensions: C/C++ Extension Pack by Microsoft, PlatformIO IDE
- Bambu Lab Slicer -
<https://bambulab.com/en-us/download/studio>
 - Printers: X1 Carbon 0.4 mm / P1S 0.4 and 0.6 mm
 - Filaments: Generic PLA, Generic PETG, Bambu PLA Basic, Bambu PETG-HF
- Git -
<https://git-scm.com/downloads>
 - Override default branch name from "master" to "main"

Software Links Continued



- Silicon Labs “CP210x Driver”
<https://www.silabs.com/developer-tools/usb-to-uart-bridge-vcp-drivers?tab=downloads>
- Raspberry Pi Imager -
<https://www.raspberrypi.com/software/>
- Google Drive for Desktop - (At the bottom of the page under Sync your desktop with Drive)
<https://workspace.google.com/products/drive/#download>
- LinkedIn Learning Activation - (Not software but free resource)
<https://linkedinlearning.tamu.edu/>
- Github Student Developer Pack - (Not software but free resource)
https://education.github.com/discount_requests/application



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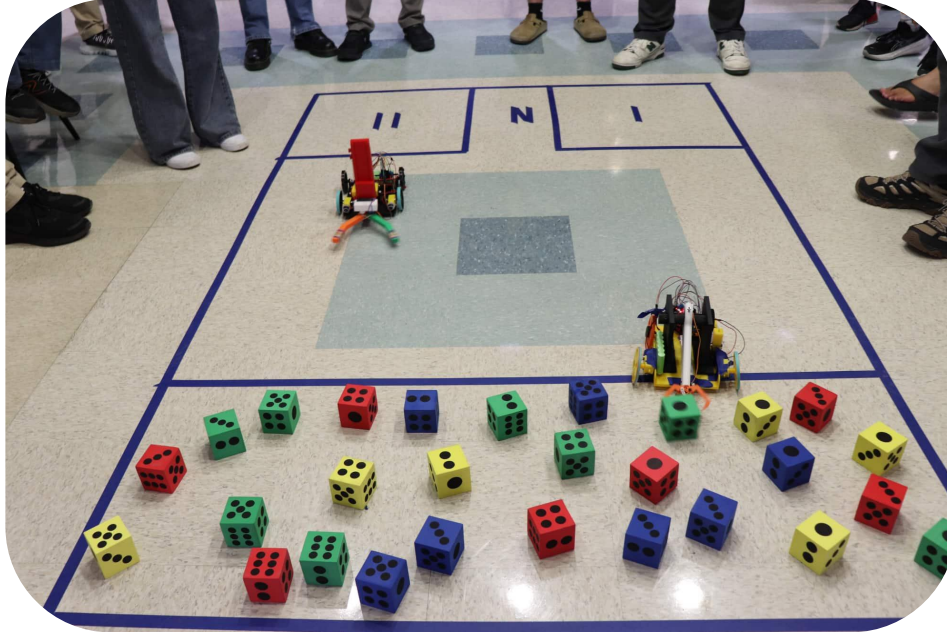
Project Reveal



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TURTLE Towers

Introduced: Spring 2025



Records:
Tallest Stack: 6
Most Blocks: 16



Story

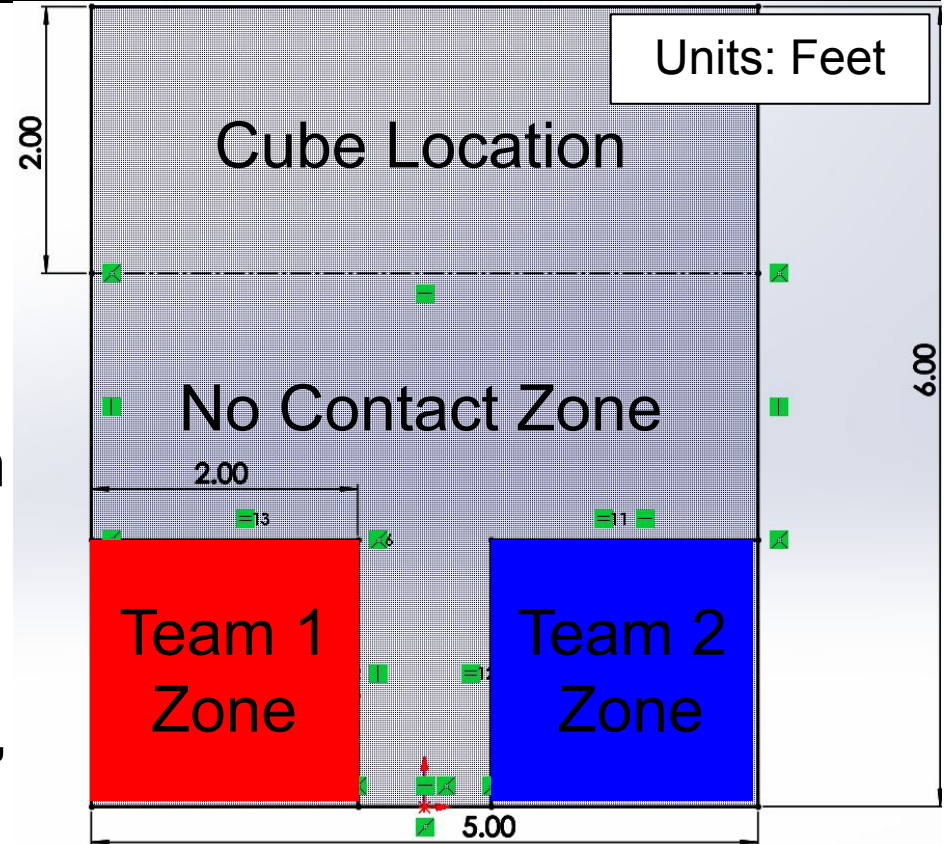


Squeeze Enterprises has recently acquired rights to build student housing in College Station. The contractor chosen to develop their properties will be whomever can acquire the most points in two and a half minutes. The contract states that points are awarded on the number of living areas (blocks) and the height of the tallest skyscraper. You are contractors trying to win the lucrative contract. May the best contractors' robot win.

Gameplay



- 2:30 Minute Match
- Robots begin in their respective “Team Zone”
- Robots will traverse to the “Cube Location” and bring blocks back to their “Team Zone”
- Building blocks are randomly distributed within the “Cube Location”



Points



- 1 Point per block in Team Zone
- 12 Points per stack level (Only on highest stack greater than 1)
 - Stack: Extra Points -> 1:0, 2:12, 3:24, 4:36, 5:48
 - Stack must be vertical with bottom face less than 1 mm above the ground
- 5 Points if robot integrates sensor data
 - Turning on an LED light
 - Sending sensor data to the controller
 - Automating a task
 - etc
- 2 Points if your robot doesn't rely on adhesives, zip ties, or similar assembly securement solutions
 - Sometimes these solutions are the best option. Please provide justification for these exceptions.
- All BONUS points (Sensor/no adhesives) are awarded once your first block is scored.

Rules



- Building Blocks are 2.5 in x 2.5 in x 2.5 in
- You may not collide/interfere with an opposing team in the “No Contact Zone” or enter the opposing “Team Zone”
 - Priority goes to robots returning to their “Team Zone”
- Points are scored at the end of the match
 - Blocks that enter into opposing “Team Zone” are scored for the opposing team
- Blocks must be completely within the “Team Zone”
 - “Team Zone” includes the tape
- Good Sportsmanship Rules
 - Play fair and be respectful



Pay your dues!!!

Links should be in your orgs communication platform of choice (Discord)

Must be Paid by September 25th

Make sure you are in your orgs **Discord** and nicknamed
“First Last” → Very important to not get kicked from the
Discord

(Also turn on notifications for the announcements channel)

TURTLE Welcome Social



**September 17th at 6:00 PM
AT SIMPSON DRILL FIELD**

Drinks and popsicles provided!!!





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SolidWorks (CAD) Foundation

Next Week
Bring a computer mouse and make sure Solidworks
is installed!



**“If you only do what you can do, you'll never
be better than what you are.”**

Master Shifu (Kung Fu Panda 3)



Hatchling